EASTERN MICHIGAN UNIVERSITY	IA-427 Forensics Report Internal Case Dante David Incident Status: **In progress**	Report written by: Julio Barros Revision: 1

Read the document carefully – I have removed some required sections in a full report and streamlined others. I've done this to reduce your assignment time but keep the main components. Every section that is highlighted needs your input. When outlining your artifacts, include the elements covered in class. You may or may not find all artifacts, but keep the report in artifact order by computer image and USB image (see below)

1. Download "Dante David Case\Ddavid_computer_image.zip" and "DDavid Memdump" share;
2. FYI - You could install any tools (Autopsy, WRR, hex tool, etc.) you want to use directly on your OS and save the overhead of a virtual machine.

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Executive Summary

Dante David, an employee at ABC Incorporated, is being looked for selling drugs using corporate resources. Normally, corporate investigators would not handle a criminal case, but we'll move forward in this case. You have been provided with an image of the David computer and active RAM.

Things to be aware of for your investigation:

1) Any images of drugs or drug-related material are considered evidence in this case. Review all artifacts using the same methodology done in the Audrey Simmons case.

2) All artifacts should be analyzed and in your report. They may contain evidence of the case in the format of file signature changes, steganography, or other forms of suspect tampering.
3) There are eleven artifacts in the system image and USB image. Artifacts are numbered 1-11 and must appear in order on your report based on the top-level artifact. By this, I mean that when you find (example) Artifact 4, and then discover Artifact 2 and Artifact 8 are hidden inside, your report will be ordered by Artifact 4.
    a. Artifact 4
        i. Artifact 2
        ii. Artifact 8

4) Ignore date and time stamps in your investigation
5) A rubric for the exam can be located in Canvas

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4. Results and Findings (follow this outline in your report. Report as we've done in previous full cases)

4.1 **What artifacts are recoverable from the computer image? Comments?**

- Artifact #2 - coke1.png (Was found on the DDavid Pictures files.)



- Artifact 4 – drugs1.jpg (was a zip file encrypted inside the file coke1.png)



- Artifact 9 – Drugs6.jpg (Also found on the zip file inside the coke1.jpg)



- Artifact 10 – Order Details.xlsx (Also on the zip file on the coke1.png)

Artitifact 10

Client	item	cost	paid
Jeff	Cocaine	$ 175	Y
Greg	Weed	$ 45	N
Darla	Oxy	$ 215	N
Fred	Meth	$ 75	Y
		$ 510	

- Artifact 3 - Coke2.jpg (This artifact was found on the DDavid Downloads files, the user deleted this artifact)



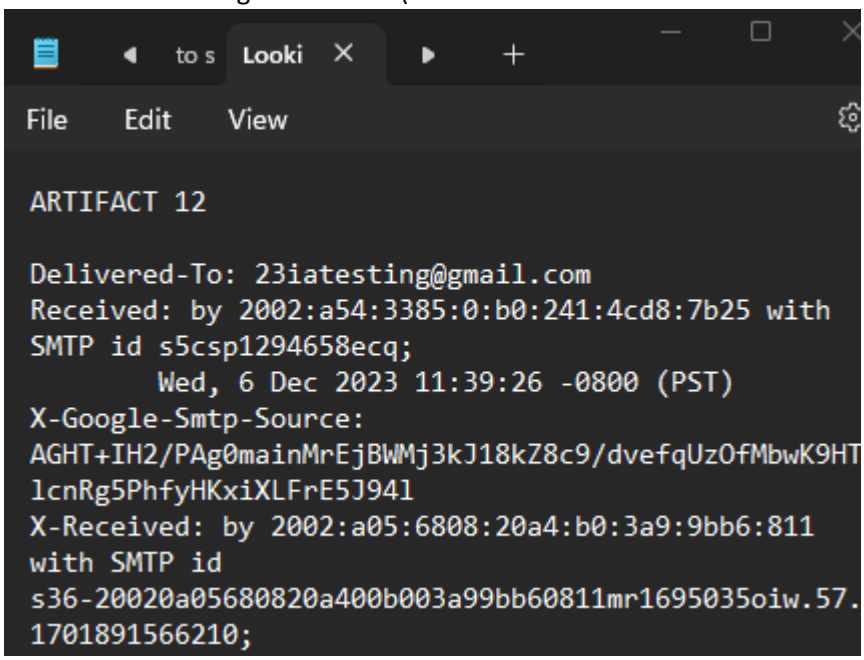
- Artifact 11 - Pepsi.jpg (This artifact was found on the DDavid Desktop)



  - Artifact 8 – Pepsi.jpg (This artifact was crypted in the pepsi artifact, used the tool HexEd to find it)



- Artifact 12 – Looking to score.txt (was found on the DDavid documents file)



  - Artifact 5 – Looking to score.txt (Was found inside the txt using the email forensics technique.)



  - Artifact 6 – Looking for score.txt (Also found inside the txt file using the email forensics technique)

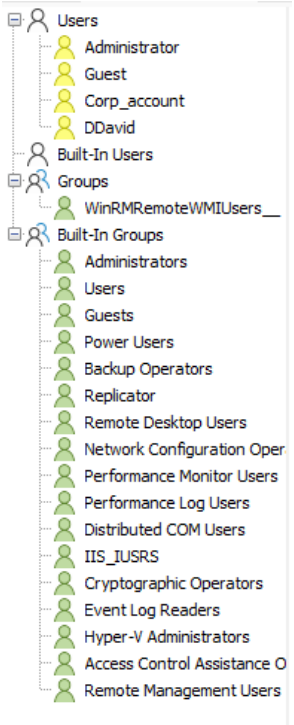


4.4 **Does a review of the system image confirm that anti-forensic techniques were used on the Simons computer?**

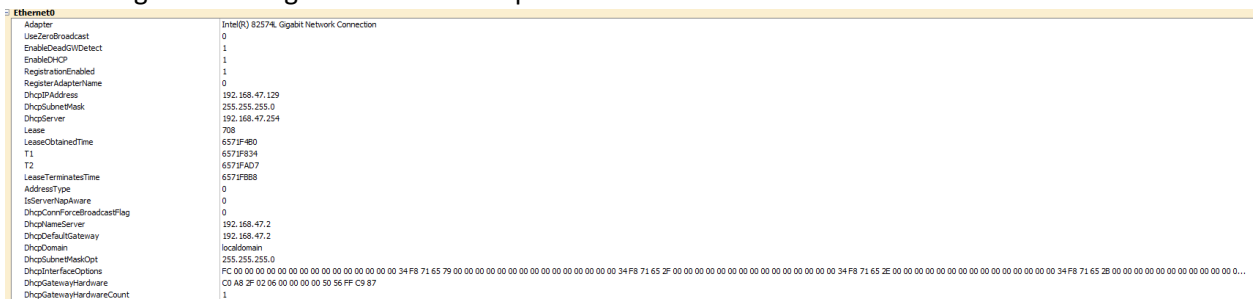
- Mr. Dantes used couple techniques on the computer to be able to sell his drugs using the enterprise machine, Encryption, Overwriting data in some files and Steganography were the techniques that I could find reviewing his machine.

4.5 registry

- Screengrab the user accounts on the image



- Screengrab the assigned IP on the computer

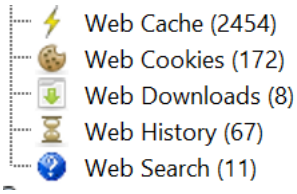


Ethernet0	
Adapter	Intel(R) 82574L Gigabit Network Connection
UseZeroBroadcast	0
EnableDeadGWDetect	1
EnableDHCP	1
RegistrationEnabled	1
RegisterAdapterName	0
DhcpIPAddress	192.168.47.129
DhcpSubnetMask	255.255.255.0
DhcpServer	192.168.47.254
Lease	708
LeaseObtainedTime	6571F4B0
T1	6571F834
T2	6571FAD7
LeaseTerminatesTime	6571FBB8
AddressType	0
IsServerNapAware	0
DhcpConnForceBroadcastFlag	0
DhcpNameServer	192.168.47.2
DhcpDefaultGateway	192.168.47.2
DhcpDomain	localdomain
DhcpSubnetMaskOpt	255.255.255.0
DhcpInterfaceOptions	FC 00 00 00 00 00 00 00 00 00 00 00 00 00 00 34 F8 71 65 79 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 34 F8 71 65 2F 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 34 F8 71 65 2E 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 34 F8 71 65 2B 00 00 00 00 00 00 00 00 00 00 00 0...
DhcpGatewayHardware	C0 A8 2F 02 06 00 00 00 00 50 56 FF C9 87
DhcpGatewayHardwareCount	1

4.6 Browser forensics - *note I received a few emails that mentioned browser forensics are a little fuzzy. With that in mind, I made the case requirements lighter and provided guidelines for some of the material.

Mr. David was running Chrome and Edge has his browsers. Edge was actively running on the computer, and you'll see evidence of that in your RAM analysis. Mr. Davis was looking at a hacking tool called "rubber ducky" and considering fleeing to a foreign country without extradition to the US. Find evidence of these actions and provide comments and supporting screengrabs. Mr. Davis was able to delete his Chrome history before we imaged his computer, but his Edge data remains intact.

Some help: Edge history can be found using the tools from the lab, or directly in Autopsy. In Autopsy, review the following locations:



Is there any evidence in Edge for this case? As you can see in the Edge Web Search you can find very important information, such as how to use ruberducky which is an USB Hack. And the search about how to sell drugs online as well.

Source Name	S	C	O	Domain	Text	Program Name	Date Accessed	Data Source
History				bing.com	libre office	Microsoft Edge	2023-11-08 13:43:27 EST	DDAVID_COMPUTER_IMAGE.001
History				bing.com	libre office	Microsoft Edge	2023-11-08 13:43:27 EST	DDAVID_COMPUTER_IMAGE.001
History				bing.com	chrome download	Microsoft Edge	2023-12-06 14:57:41 EST	DDAVID_COMPUTER_IMAGE.001
History				bing.com	chrome download	Microsoft Edge	2023-12-06 14:57:41 EST	DDAVID_COMPUTER_IMAGE.001
History				bing.com	ruberducky	Microsoft Edge	2023-12-06 15:00:30 EST	DDAVID_COMPUTER_IMAGE.001
History				bing.com	ruberducky	Microsoft Edge	2023-12-06 15:00:30 EST	DDAVID_COMPUTER_IMAGE.001
History				bing.com	how to use ruberducky	Microsoft Edge	2023-12-06 15:01:18 EST	DDAVID_COMPUTER_IMAGE.001
History				bing.com	how to use ruberducky	Microsoft Edge	2023-12-06 15:01:18 EST	DDAVID_COMPUTER_IMAGE.001
History				bing.com	how to sell drugs online fast cast	Microsoft Edge	2023-12-06 15:02:51 EST	DDAVID_COMPUTER_IMAGE.001
History				bing.com	how to sell drugs online fast cast	Microsoft Edge	2023-12-06 15:03:10 EST	DDAVID_COMPUTER_IMAGE.001
History				bing.com	how to sell drugs online fast cast	Microsoft Edge	2023-12-06 15:03:10 EST	DDAVID_COMPUTER_IMAGE.001

For Chrome, I will not grade this portion as a lot of the questions from students were on deleted history. One way to review the delete history is by reviewing the Session (SSNS) and Tabs files in the Windows directory. C:\Users\<username>\AppData\Local\Google\Chrome\User Data\Default\sessions

Yes, the SSNS was deleted.

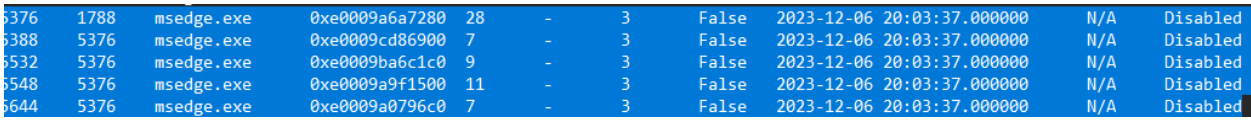
Session_13346440638241815			2023-12-07 11:38:38 EST	2023-12-07 11:38:38 EST	2023-12-07 11:37:18 EST	2023-12-07 11:37:18 EST	11255	Allocated	Allocated	unknown	/img_DDAVID_COMPUTER_IMA
Tabs_13346440639971473			2023-12-07 11:37:19 EST	2023-12-07 11:37:19 EST	2023-12-07 11:37:19 EST	2023-12-07 11:37:19 EST	44987	Unallocated	Unallocated	unknown	/img_DDAVID_COMPUTER_IMA
Tabs_13346440693314269			2023-12-07 11:38:38 EST	2023-12-07 11:38:38 EST	2023-12-07 11:38:13 EST	2023-12-07 11:38:13 EST	10989	Allocated	Allocated	unknown	/img_DDAVID_COMPUTER_IMA
[current folder]			2023-12-07 11:38:13 EST	2023-12-07 11:38:13 EST	2023-12-07 11:38:13 EST	2023-12-06 14:59:08 EST	56	Allocated	Allocated	unknown	/img_DDAVID_COMPUTER_IMA
[parent folder]			2023-12-07 11:38:38 EST	2023-12-07 11:38:38 EST	2023-12-07 11:38:38 EST	2023-12-06 14:59:08 EST	56	Allocated	Allocated	unknown	/img_DDAVID_COMPUTER_IMA

Review the files in a hex editing tool. Is there any evidence in Chrome for this case? (Comment and screengrab). Can you find other Chrome evidence in the image/registry/or ?

4.7 Memory Forensics (screengrab

1. What is the process ID for Microsoft Edge?
   5376
   5388
   5532
   5548
   5644
2. Is there evidence that the DDavid SID was running Edge?
   Yes.

3. Is there evidence that the DDavid SID was running Edge at a Mandatory Access level?
   Yes, msedge.exe runs Low, Medium, and Untrusted levels of mandatory access.
   5532 msedge.exe S-1-16-4096 Low Mandatory Level
   5548 msedge.exe S-1-16-8192 Medium Mandatory Level
   5644 msedge.exe S-1-16-0 Untrusted Mandatory Level
4. Is there evidence that Edge was running in the current session (a session would be actively running, but a process ID could be a remnant of a program that could be dormant).



5376	1788	msedge.exe	0xe0009a6a7280	28	-	3	False	2023-12-06 20:03:37.000000	N/A	Disabled
5388	5376	msedge.exe	0xe0009cd86900	7	-	3	False	2023-12-06 20:03:37.000000	N/A	Disabled
5532	5376	msedge.exe	0xe0009ba6c1c0	9	-	3	False	2023-12-06 20:03:37.000000	N/A	Disabled
5548	5376	msedge.exe	0xe0009a9f1500	11	-	3	False	2023-12-06 20:03:37.000000	N/A	Disabled
5644	5376	msedge.exe	0xe0009a0796c0	7	-	3	False	2023-12-06 20:03:37.000000	N/A	Disabled

4.8 **Conclusion**

- In conclusion, based on the comprehensive information gathered in this case, it is evident that David Dantes was indeed involved in the illicit sale of drugs utilizing corporate equipment and machines. Through this investigation, I have successfully identified records showcasing clients, products, and associated prices. Additionally, compelling evidence in the form of photographs and emails has been obtained, conclusively establishing Mr. Dantes as a drug dealer.